

Quiz - Curve Sketching

Part A: True or False – Place your answer in the space provided on the right.

1.

If $f(x)$ is a polynomial of degree 3, then its graph has one point of inflection.
2.

If $f''(1)=0$, then $(1, f(1))$ must be a point of inflection.
3.

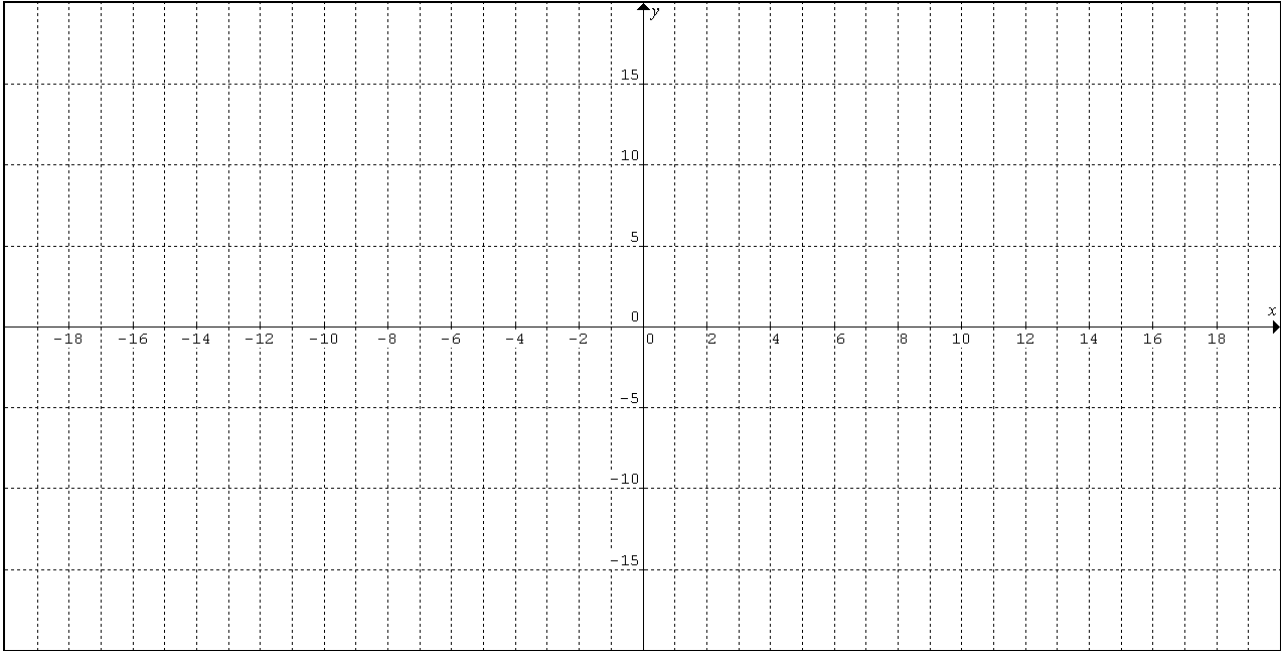
If $f'(c)=0$, then f must have either a relative minimum or a relative maximum at $x=c$.

Part B: Full Solutions

4.

Determine any local maximum or local minimum points and the intervals of increase and decrease for the function $f(x)=\frac{1}{7}x^{\frac{7}{3}}-x^{\frac{4}{3}}$.
5.

Sketch the graph of a function f such that $f(-2)=2$, $f(0)=f(2)=0$, there is a horizontal asymptote at $y=1$, there are vertical asymptotes at $x=-1$ and $x=1$, $f''(x)>0$ for $x<-1$, $f'(x)<0$ for $-1< x < 1$ and $f''(x)<0$ for $1 < x$.



6. For the function $f(x) = 2x^3 - x^4$, determine the critical points and classify them using the second derivative test.
7. The graph of the function $f(x) = \frac{ax+b}{(x-1)(x-4)}$ has a horizontal tangent line at the point (2, -1). Find a and b and show that the function has a relative maximum at (2, -1).
8. Given $f(x) = 3x^4 + x^3 - 21x^2$.
- Find the absolute minimum value of $f(x)$. Show all work.
 - Find the x -coordinate of each point of inflection on your graph of $f(x)$.

9. Consider the function $f(x)=x^{\frac{4}{3}}+4x^{\frac{1}{3}}$ that has $f'(x)=\frac{4}{3}x^{\frac{1}{3}}+\frac{4}{3}x^{-\frac{2}{3}}$ and $f''(x)=\frac{4}{9}x^{-\frac{2}{3}}-\frac{8}{9}x^{-\frac{5}{3}}$. Sketch a properly labelled graph of $f(x)$.

